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Detecting Landfill sites through YOLOv3 (Powered by Darknet) using Satellite Images

BSc Hons Computer Science

Submission Date: 25/03/2022

Declaration of Originality

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1. Abstract

Research in object detection in computer vision have grown exceptionally over last few years. The initial aim of this project was to put a machine learning model in use to track down landfill sites using ariel data collected from satellites. As environmental agencies across the world are struggling to keep up with the violations in waste management industry, it has become a necessity to have a system that can help track illegal landfill sites down to reduce environmental damage. For this purpose, machine learning technology can be leveraged to automate the process of finding landfills and waste dumps in large volume of datasets.

Deep learning-based models are prone to extract and learn unwanted information, given that inaccurate input data was provided. It is referred as underfitting or overfitting. This research outlines the architecture of YOLOv3 and how it is implemented to detect landfill sites. It also evaluates model's performance using industry standard metrics. The results obtained from the model vary due to the complexity of the target objects in dataset. All findings are concluded at the end of the research alongside suggestions that can contribute towards achieving more accurate detections.

2. Report Structure

This section highlights the structure of report and provides a brief overview of frameworks, models and various performance evaluation techniques discussed below.

- **Introduction:**
  Provides an overview on the challenges faced by environmental agencies due to illegal landfill sites and their side effects on economy and environment, while proposes an innovative solution to help reduce environmental pollution, expenses and labour being invested toward tracking down illegal landfill sites.

- **Background:**
  Highlights the technical specifications of Darknet-53 and YOLOv3 and how these two models work together to perform object detection through their multi-layer convolutional neural network. It also gives a brief overview of features and settings that can be tweaked for each dataset to achieve best possible performance and accuracy.

- **Dataset Preparation:**
  Outlines the tools and techniques used to collect high quality satellite imagery while also emphasizing on quality of annotation that is required for supervised learning. Model's accuracy is directly related to the quality of annotations.

- **Model Training:**
  This section of report demonstrates the custom configurations made to the model to train it on the type of dataset collected for this project. Different type of network image sizes and their effects on prediction outputs are discussed.

- **Analysis:**
  This section uses evaluation metrics to highlight the performance and accuracy of the YOLOv3 model and gives a brief explanation about challenges faced during training and testing of the model.

- **Conclusion:**
  This section concludes the research and implementation done for the aims set for this project. Highlights some crucial findings from the project and provides overall feedback on factors that can help and improve the model's accuracy for predictions on satellite imagery.

3. Introduction

3.1. Motivation

Landfills play an essential role in reducing air pollution, greenhouse gases, and water pollution. These sites are mainly located far away from residential areas, which significantly improves air quality near populated areas. Local councils manage these sites in their respective cities. Despite all the restrictions and regulations, illegal dump and fly-tipping sites have been appearing in some commercial regions and locations further away from populated areas as they are unchecked and visited less frequently. These illegal dumpsites create an environmental and health hazard for all inhabitants in the region as the waste is not recycled or contained in a specific containment. Governmental bodies have been trying to track and take down these sites but faced countless challenges: finding an efficient way to track such sites in the shortest time is one.

Between 2018 and 2019, Britain's environment agency disclosed the cost for compensating the damages caused by such activities to be around £924 Million; there has been an increase of 90% in damages since 2015 **[1]**. These costs get deduced from the contributions made to the economy by taxpayers. The government has attempted to leverage technology to overcome the tracking obstacle by using a countrywide CCTV network and ariel data collected from drones and satellites, but the entire process turned out to be time-consuming while requiring human input. To automate and speed up the whole dumpsite tracking process, machine learning models can come to use. Machine learning models programmed to classify objects in images can identify landfills in the significantly more extensive dataset at a higher pace. With the availability of free satellite imagery from trusted sources such as NASA, Google, and Sentinel Hub, in-person visits are no longer required to capture visual data for suspected regions.

3.2. Scope of the study

This project aims to generate a dataset of high-resolution satellite imagery. This process includes obtaining the images and annotating the dataset so that the deep learning model can adapt and learn to identify the target object, which is a waste dump in this case. The deep learning model will be fed a dataset with slight variations to observe and evaluate results. The deep learning multi-layer neural network studied and implemented for this research is You Only Look Once v3 (YOLOv3). In depth study and analysis of this model will be taken in order to evaluate its strengths in regard to object detection, image classification and localization.

This research will also investigate transfer learning techniques and implementation for the YOLOv3 model. This feature enables the same CNN-based model to be trained across various datasets using same the weights obtained from the previous training session.

4. Background

4.1. Darknet-53

Darknet is a neural network framework for image classification and object detection. It is written in C and CUDA, allowing the model to perform classification and object detection precisely at incredible speed. It is the backbone of YOLOv3; hence, it powers the model using its custom 53 convolutional layers trained on datasets of millions of images on ImageNet. Darknet's 53 convolutional layers extract the features present in the given dataset. NVIDIA's CUDA framework has been extensively utilised in the development of Darknet; hence, it fully utilises the GPU and computes incredible scores marginally higher than other competitors. Darknet-53 is well optimized even with its 53 convolutional layers that it can compute highest floating-point operations per second (BFLOP/s) as show in comparison table below **[2]**. BFLOP/s benchmarks computational capability of a system or framework.

Backbone	Top-1	Top-5	Ops	BFLOP/s	FPS
Darknet-19	74.1	91.8	7.29	1246	**171**
ResNet-101	77.1	93.7	19.7	1039	53
ResNet-152	**77.6**	**93.8**	29.4	1090	37
Darknet-53	77.2	**93.8**	18.7	**1457**	78

Figure (1) – Comparison between well-known backbones. BFLOP/s refers to billion floating-point operations per second, FPS refers to frames per second, and OPS refers to billions of operations. Source **[2]**.

4.2. YOLOv3

YOLOv3 was developed on top of Darknet. YOLOv3 comes with additional 53 convolutional layers that total the layer count to 106 for the model as shown below in Figure (2). YOLOv3 provides a significant performance advantage over its predecessors, YOLO & YOLOv2, while also offering state of the art features such as residual block, up-sampling, and Intersection over Union (IoU) **[2]**.

This model uses additional convolutional layers rather than pooling layers to down sample the images after every ten iterations. The down sampling factor is a multiple of 32; hence, for default training configuration, it typically scales input network size between 448x448, 608x608 and 480x480 Etc. The down/upscaled images get passed into each layer while traversing through the network. These additional convolutional layers avoid the loss of low-level features while down sampling a feature map; otherwise, such low-level features are excluded by pooling layers in traditional CNN-based machine learning models. YOLOv3's only three layers support object detection, as shown in Figure (2).

Its architecture relies heavily on auto-scaling and splitting the image at a given layer into grid cells, then assigning a bounding box for each grid cell detected in the centre of the target object. This model performs predictions at three scales: layers 82, 94 and 106 by down sampling the images by stride numbers 32, 16 and 8 for each layer. For example, an image with an input size of 416x416 passes through a 32-stride layer is down sampled to 13x13 size while ensuring symmetrical dimensions. Likewise, the rest of the stride layers behave similarly while down sampling the images; therefore, the network input size needs to be a multiple of 32 **[4]**. Feature maps computed from down sampling are of similar dimensions as well.

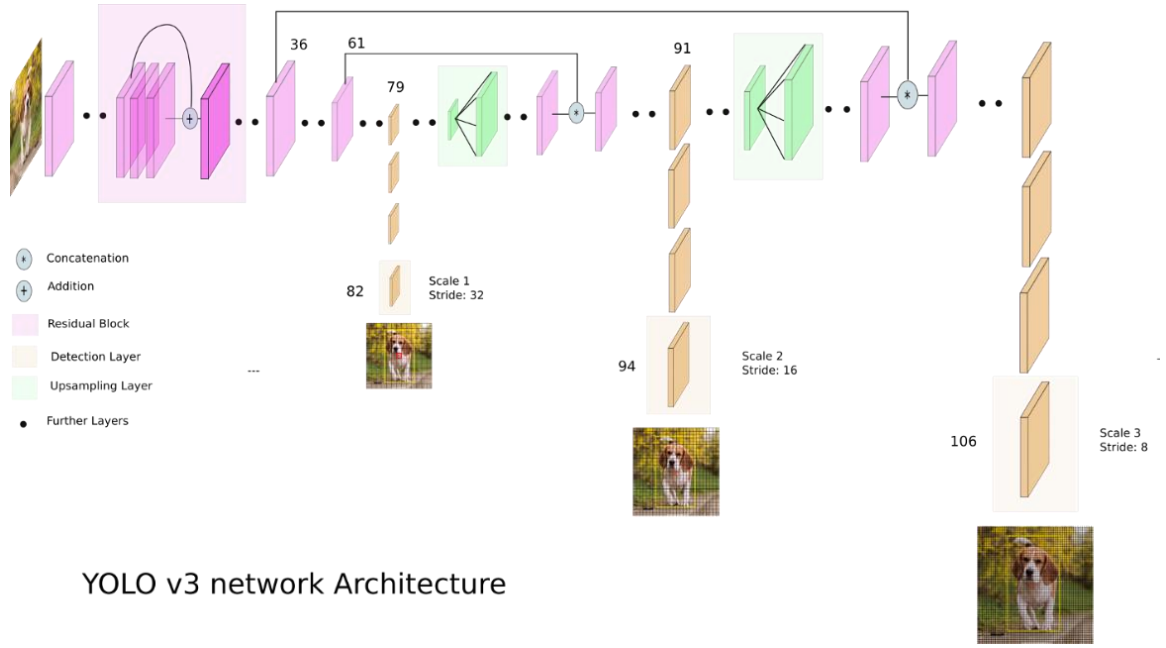

Figure (2), Source - [3]

YOLOv3's scale-1 best detects objects marginally bigger than other objects in the same image, while scale-3 performs best predictions for small objects. Scale-2 lies between 1 & 3; however, this model struggles to detect small objects in specific datasets. Satellite imagery is no exception as often target objects are hardly distinguishable due to poor image quality, camera angle or lighting.

4.2.1. Object Detection

YOLOv3 outperforms its predecessors in object detection by taking advantage of multi-label classification i.e., by making a single object member of multiple classes. The backbone of this model, i.e., Darknet-53, is written in C, allowing the model to perform classification and object detection precisely at incredible speed. YOLOv3 uses a regression-based algorithm that computes confidence level for bounding boxes for a whole image in one go. The detection kernel for YOLOv3 has a structure of 1x1 and represents feature maps at different

detection layers such as 13x13, 26x26 and 52x52. These feature maps are obtained through down sampling of images. The depth of the detection kernel is represented by:

$$Depth = (B * (5 + C))$$

"B" refers to the number of bounding boxes per grid cell. By default, YOLOv3 has three bounding boxes per grid cell and "C" refers to the number of classes present in the dataset. Feature maps inside the network are dependant on this depth attribute as it houses core information about the dataset and the model. An example of feature map can be given as *(13, 13, depth)*. Once all these attributes are obtained by a grid cell, it computes the probability of finding target object and assigns a weight value for that cell using logistic regression **[2]**.

A grid cell located at the centre of the target object returns highest weight value hence, instructing the model to set the cell as the centre point for the bounding box for the object. During training, the model keeps predicting the score for each bounding box while also offsetting the centre points until it finds a best match between predicted bounding box and the ground truth box using non-max suppression. Figure (3) illustrates the class prediction process for YOLOv3 model.

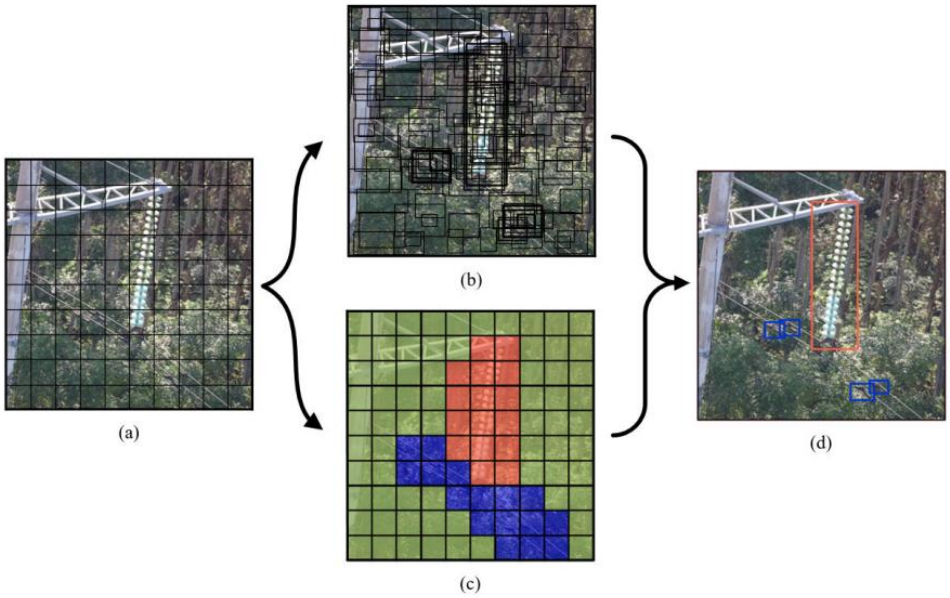

Figure (3) – Shows flowchart for bounding box predictions. Image is split into symmetric grid cells, then each gird cells computes the probability for confidence score. Finally assigns a bounding box for the grid cell with highest confidence and IoU score. Source – **[9]**

YOLOv3 uses non-max suppression technique on overlapping bounding boxes to find the box with confidence score higher than defined threshold which also covers the target object. This filters out boxes with lower confidence but does not remove the overlapping boxes that have higher confidence level. As final output should only consist of one box per object, these overlapping boxes are removed with the help of Intersection over Union (IoU). It is an evaluation matric used to assess the accuracy of the object detection convolutional layers. It

is measured by calculating the area of two boxes that are overlapping each other **[9]**. It can be computed using:

$$Intersection\ of\ Union = \frac{Area\ of\ overlap}{Area\ of\ Union}$$

$$Area\ of\ overlap = Intersection\ between\ predicted\ box\ and\ ground\ truth\ box$$

$$Area\ of\ union = Union\ between\ predicted\ box\ and\ ground\ truth\ box$$

Like confidence score threshold, a threshold for IoU is set and a box with IoU lower than this threshold is removed from the detections. Figure (6) also demonstrates a great application of non-max suppression technique alongside IoU.

4.2.2. Improvement in Accuracy

As YOLOv3 only computes each image once for an iteration, the resultant probability of missing small objects in a high-resolution image is observed to be relatively higher. YOLOv3 can be configured to compute images using different image sizes for a set of iterations to counteract this issue. By default, the image size is 416x416 for three red, blue, and green channels. A higher multiple of 32 provides better classification results but will result in longer training times. This feature of YOLOv3 allows images with variable sizes to be trained without tiling or splitting the image into multiple sections and eradicates the need to manually handle verity of aspect ratios for the images.

4.2.3. Transfer Learning

Transfer learning have reshaped the concept of training fresh out of the box models. Machine learning models are greedy when it comes to training data. Preparing huge amount of labelled data for the model is a slow and lengthy process and demands extensive human input. To save time and resources, such models can leverage pre-trained weights from other trained deep learning models that have seen similar or different dataset before such as having knowledge about shadows, colour grading and daily use objects **[5]**. This significantly reduces the training time for the new model through supervised learning. YOLOv3 is very much capable of using pre-trained weights to train on a new dataset.

4.3. Dependencies

This project makes use of open-source tools and libraries. As Darknet is developed using CUDA, any modern NVIDIA GPU that supports CUDA can run this framework with exceptional performance results. It also relies on CUDA Deep Neural Network Library (cuDNN) for GPU acceleration. Darknet is compiled using OpenCV (Version < 4.0) and CMake **[8]**. Furthermore, for testing purposes, few other required libraries are as “sklearn” and “numpy”.

5. Dataset Preparation

5.1. Image collection

Dataset is just as important as any machine learning model, especially in the case of supervised learning. Good quality and a well-labelled dataset are necessary for a machine learning model to detect patterns from the input and then compute predictions on unseen data. Obtaining a large volume of a high-quality dataset is a hectic process that poses many challenges during the collection process. A dataset should not hold too many negative samples as this could severely affect the class prediction capabilities of the machine learning model, while training a model on a nearly ideal dataset might result in overfitting. A typical dataset is usually split into training and testing data. The testing dataset is used to validate the model's object or class prediction capabilities; this is only possible after training the model on the training dataset. In the case of supervised learning, labelling the training dataset is crucial and necessary. Training dataset must be carefully prepared to keep factors minimum that can compromise the model's performance and prediction abilities.

This project aims to train the YOLOv3 model to detect landfill sites from given test data. As YOLOv3 supports input datasets with variable image sizes, input images do not have to be resized or cropped beforehand hence saving us a reasonable amount of time and labour. YOLOv3 offers excellent flexibility in image size and aspect ratio as it samples up and down images when required. On the other side, satellite imagery of lower resolution can affect the model's ability to propose predictions for the objects. Smaller waste dumps can lose their semantic value when images are sampled down in the convolutional layer while training. Hence, if the dataset contains too many such samples, the model might mistake the waste dumps as typical terrain, given that most landfills are located far away from populated areas and surrounded by empty or vegetated fields.

While keeping all those factors in mind, Google Earth was used to collect images of up to 4K resolution. Google Earth holds worldwide satellite data of tens of petabytes obtained from base map service providers such as Landsat, Sentinel satellites, NSW Health, and US Geological Survey **[6]**. It also offers to view and download satellite imagery from previous years. This feature sped up the data collection process significantly as multiple samples of the same landfill site were taken over 4 to 5 years. Viewing and analysing the imagery on the go was highly convenient as numerous landfill sites either had no target object or had waste stored in containments; this significantly helped avoid underfitting of the model as the ratio of negative to positive samples was much lower.

Figure (4) - Images collected from Google Earth

(Difference in waste dumps in a developed & non developed country)

Most images are collected from the UK, USA, Canada, South Korea, and China. These developed countries have well maintained and regulated landfill sites, but this led to the issue of having fewer target objects in the images. A good volume of images is collected from developing countries such as Pakistan, Brazil, Nigeria, and India to counteract this issue. The images collected from developing countries provided a healthy amount of required target samples.

5.2. Annotation

The dataset's quality has a crucial impact on the output of a machine learning model. Image annotation refers to labelling target objects in each image in a dataset so that the model can extract the features, adapt, and learn the pattern. During annotation, each object gets assigned a unique class name. YOLO family have got its annotation format. The model requires a text file named similar to its corresponding image name. The format for the label file for YOLO is as follow:

<object-class> <x> <y> <width> <height>

Every class has a unique numerical identifier which is represented by “object-class”. The centre point of a bounding box is represented by 2D coordinates “x” and “y”, while “width” and “height” refer to the size of the bounding box. For this project, a GUI based open-source annotation tool "LabelImg" has been used [7]. It offers basic functionality to annotate images for multiple classes and stores them in YOLO format. Below is an example of what an annotation file looks for an image with two classes:

0 0.565937 0.581311 0.031458 0.048219
1 0.603437 0.537819 0.045208 0.068390

Landfill sites hold various types of waste. It can vary from recyclable household waste to toxic industrial waste. Waste dumps differ in size, shape, and height. The satellite images can observe a very subtle pattern in waste dumps. Unlike other objects, waste dumps can be challenging to detect using a machine learning model as they can blend in with surrounding vegetation or mud. To make it easier for the model to learn the pattern in shapes, the boxes need to be drawn slightly further away from the borders of waste dumps. It ensures that a borderline between ground and waste dump is visible.

Figure (5) - Label on left side helped model learn about waste dumps quicker and easily as compared to the one on right.

Initially, majority of the waste piles were labelled wrong which generated some interesting results. With border not being visible, model could not differentiate between a waste pile and neighbouring mud pile or ground. The model would predict the whole area to be a waste pile. The waste dumps were being predicted correctly as the grid cell in the centre of the object generated correct scores, but it was not close to ideal output. This result proved the importance of refined and high-quality dataset while training any model. Hence, the annotations for all the images were updated and repolished again with this time, keeping the borders of waste dumps inside the labelled boxes as it can be seen in the image above. This rework on annotations lead to significant change in the class predictions.

6. Model training

This project used pre-trained convolutional weights for Darknet-53 trained initially on ImageNet. Even though the dataset for this research differs from the one on which the weights are trained, it gives the model sufficient feature and class knowledge to start processing the satellite images. The use of transfer learning significantly reduced the training time of the model. YOLOv3 model used in this research is trained on Google Colab using NVIDIA TESLA P100 GPU. Darknet comes with predefined configurations to train YOLOv3, but these configurations have been changed to optimise performance for the satellite dataset.

YOLOv3 trained using an initial network image size of 416x416 with a maximum batch size of 4000. It used a random coefficient value to choose between variable image sizes during training. After every ten iterations, the network up and downscaled the images to a randomly selected resolution, that is, a multiple of 32. Hence, convolutional layers of the model can constantly extract features at different resolutions, resultantly increasing class prediction accuracy. The size of "max_batches" for the model depends on the number of classes present in the dataset. It is calculated using:

$$max_batches = 2000 * classes$$

Developers of Darknet suggest setting a minimum max_batch size of 4000 even if the dataset contains only one class. Darknet generalises the training dataset by splitting images into a number of batches. By default, it loads 64 images per iteration and splits this batch of 64 into 16 mini-batches. It significantly decreases training times while utilising the most available GPU memory. If used with a higher network image size, this configuration can cause the model to crash due to insufficient graphics memory.

Figure (6) – Image on left shows bounding boxes predicted by the grid cells at 200 iterations

Image on right shows prediction made by the model after 2000 iterations with the help and non-max suppression and IoU.

During training, different network image sizes were evaluated. Manually setting the network image size larger than 416x416 resulted in crashes due to memory issues. As YOLOv3 tends to detect small objects at higher resolution, this memory issue limited the model's potential. Hence, the model is configured to switch between variable network image sizes after every 10th iteration, to counteract the issue.

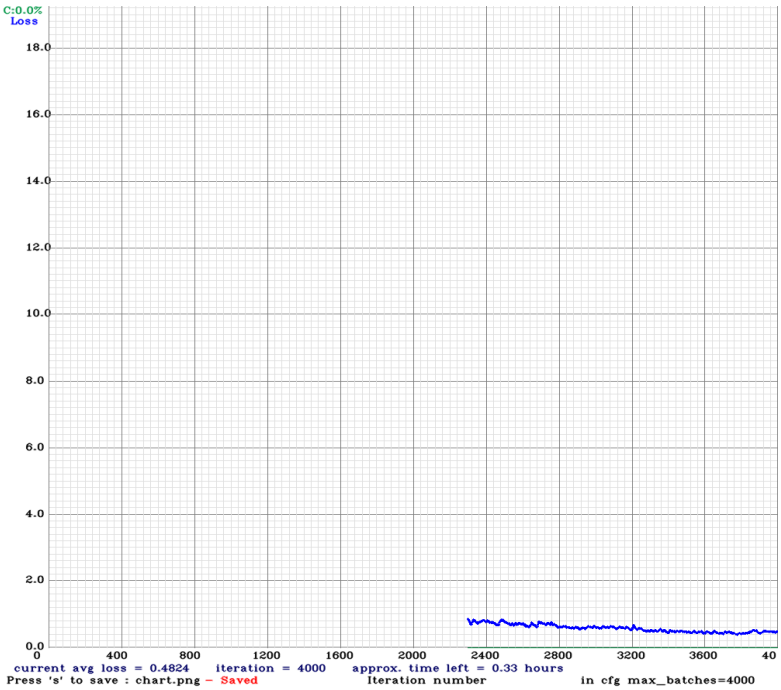

Figure (7) – Average loss calculated at training time

This configuration resulted in a much quicker training time for the complete dataset due to the smaller network image size. Figure (7) shows the graph for average loss in predictions made by the detection layers at three different scales between iteration 2300 to 4000. The whole projection of average loss value could not be saved as the model overwrote the graph due to an unexpected restart.

7. Analysis

This section of the report highlights the critical evaluation of the YOLOv3 model that is trained on satellite images. Furthermore, certain factors that encouraged underfitting and overfitting for the dataset are discussed as well. The model has been evaluated on validation dataset which does not belong to training dataset. During data collection, multiple samples for same landfill site were taken to see how well the model generalises the classes and performs under various scenarios such as testing an image of landfill (without waste dumps) that was used in training dataset

7.1. Evaluation Metric

A machine learning model can be tested on various evaluation metrics. These can highlight the strengths and weaknesses of any model numerically. For this project, "Precision" is used to evaluate the accuracy of the model regarding object detection and avoiding detecting unwanted objects such as mud pile, vegetation or other various objects that can overshadow actual waste dumps. Precision is evaluated using:

$$\text{Precision} = \frac{True\ Positives}{True\ Postives + False\ Positives}$$

"True positives" is a type of classification metric that defines the outcome of the object detection process. When a prediction is made by the convolutional layers of YOLOv3 and the bounding boxes are filtered out through non-max suppression and IoU, turns out to be part of correct class. False Positive refers to a prediction of object that is detected by the model but does not belong to any class in the dataset.

Type of Dataset	Precision	Threshold	YOLOv3 - Iterations
Training Dataset (50%)	76%	> 0.5	4000
Validation Dataset	65%	> 0.5	4000
Validation Dataset (Contains images of same landfill sites but with less waste dumps)	60%	> 0.5	4000

Figure (8) – Shows a comparison in precision values calculated by testing the model against different type of datasets.

Another metric for a machine learning model's evaluation is Mean Average Precision. Mean of average precision for the classes present in the dataset for a specified IoU threshold gives Mean Average Precision (mAP). Figure shown below shows different calculations made by Darknet-53 for the training dataset.

```
 calculation mAP (mean average precision)...
 Detection layer: 82 - type = 28
 Detection layer: 94 - type = 28
 Detection layer: 106 - type = 28
664
 detections_count = 3848, unique_truth_count = 1756
class_id = 0, name = waste, ap = 91.26%         (TP = 1638, FP = 210)

 for conf_thresh = 0.25, precision = 0.89, recall = 0.93, F1-score = 0.91
 for conf_thresh = 0.25, TP = 1638, FP = 210, FN = 118, average IoU = 69.05 %

 IoU threshold = 50 %, used Area-Under-Curve for each unique Recall
 mean average precision (mAP@0.50) = 0.912625, or 91.26 %
Total Detection Time: 163 Seconds
```

Figure (9) – mAP calculated by Darknet-53 on our dataset

It returned a Mean Average Precision of "0.91" for IoU threshold of "0.5". The model observed 1638 True Positives that means that 1638 predicted bounding boxes were matched to original ground truth boxes. This result can vary as we move from lower to higher iterations for trained weights.

7.2. Challenges in waste dump detection

Satellite imagery collected from Google Earth contains various type of waste dumps that differ by colour, height, object size, and size of pile. These attributes of any object contribute towards its uniqueness, and these are what make each class distinguishable from others. YOLOv3 model trained on the dataset have been observed to struggle under bright lighting and reflections. Light reflected from bright objects can project an image similar to a pile of waste when an image is down sampled to network image size of 416x416 or lower.

Figure (10) – Image on left, electricity towers detected as waste dump due to their angle, lighting, and subtle similarities in their pattern. Cropped Image on right is the same image down sampled to 416x416.

Figure (10) shows that the model mistook an electricity tower for a waste pile due to the similarity in their appearance at low resolution. While YOLOv3 is famous for correct predictions in case of small objects, image quality can significantly compromise its accuracy. The computation result shown in Figure (10) is one of few samples where False Negative classification prediction has been made.

7.3. Network Image Size

YOLOv3 model trained for this project is configured to down sample input images to a default 416x416 image size. Most images in the dataset contain waste dumps of larger areas, for which, YOLOv3's 416x416 is ideal network size due to its capability to detect large objects. It also helps avoid underfitting of the model as the model does not predicts high confidence score for pile of stones or mud. While testing, the images are down sampled to 800x800 instead, this is done to allow the model to make predictions for marginally smaller objects. This way, the model makes predictions for small dumps by following the features extracted from the bigger dumps during training. Regardless of the size of objects in pictures, a low-quality input image can affect model's accuracy negatively.

8. Conclusion

YOLOv3 is notoriously fast model as compared to other industry standard object detection models as it traverses through the dataset while only performing computations on each image once, every iteration. The model used in this project succeeded to learn patterns and features for waste dumps while also showed corner cases of both overfitting and underfitting. Even though, YOLOv3 offers great flexibility with input image sizes in order to compute predictions for objects of various sizes at three different scales in its convolutional layers, an input image of lower resolution containing target object in relatively smaller size can be hard to detect. Strict use of well annotated high-quality satellite imagery can lead to excellent results. A well annotated dataset plays crucial role towards mode's performance and accuracy.

The model has met the initial aim of this project while also showing room for improvements in various sectors. For certain samples of low quality, it manages to find features of a waste dump and performed correct predictions. The memory issues faced during the training can be avoid by using more capable hardware. This would certainly allow the model to extract more features of small objects at higher network image size.

As collecting and annotating satellite imagery is a hectic process, the trained weights obtained from this model can provide a good learning foundation for a new model to extract and adapt features for a similar dataset. Transfer learning in computer vision has changed the way deep learning neural networks are trained. Being able to extract information from pre-trained weights rather than starting from scratch by using random input, remarkably reduces training times and costs towards collecting more data. Analysis of performance and accuracy for the model are evaluated and gave good insight into weak points contributing towards inaccuracies in predictions.

The idea of utilising a machine learning model to track down illegal landfill sites is very much practical and findings from the project clearly indicate the potential of such model to help fight waste crime while avoiding both developed and undeveloped countries from investing valuable financial resources into unnecessary labour.

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Appendix

Project Proposal

Project Proposal

This project focuses on object detection using Convolutional Neural Networks on images collected through surveillance drones and satellites to stop and prevent waste crime. These networks use multiple algorithms to help train the model to distinguish between irrelevant and target objects.

Initial stages of project will consist of studying and understanding fundamental techniques in image processing then project can move onto data collection and sample processing will be initialised. This project will lead to develop such AI model which will be able to help prevent environmental damage and cost towards fighting waste crime.

Introduction

Deep learning neural networks in computer vision have evolved significantly in this century. The goal of object detection is to capture the key components in a picture to identify and distinguish them from each other just as any human being would detection different objects in a frame. Object Detection has gained quite a lot of fame in various fields as neural networks can be trained to process thousands of samples to complete the required task. British law enforcement agencies have been tackling waste crime for more than a year now and to prevent country's economy from paying towards repairing both infrastructural and environment damages **[1]**.

Satellite and drone imagery can prove to be significantly helpful in this regard as high-risk areas can be monitored closely by constantly monitoring captured data. But capturing big amount of data comes with its own challenges such as manually finding traces of waste in the images of large collection can be a hectic and time-consuming process. Image processing using deep learning comes handy in this situation as deep learning models can be trained to trace waste efficiently in less time.

This method has its own challenges, drones can capture poor quality/blurry images due to instability caused by sudden change in altitude during shutter capture time, turbulence, pitch or roll of the UAV. A drone with high quality camera would be preferable to achieve more accurate results as the final images would be in high resolution hence providing more information about the objects captured. Higher amount of detail in higher resolution pictures also increase risk of getting bad output during training while also taking much longer to finish each sample.

This report gives an insight into a deep learning algorithm used in computer vision that will be used in this project i.e., YOLO. This algorithm is known for its accurate and fast object

detection capabilities in real time. This is done by assigning boundary boxes for all potential objects and calculating the probability class through Convolutional Neural Network (CNN). **[2]**.

Background

This section of report will cover the in-depth research done so far regarding satellite and drone captured image processing. Drones have been in use for surveillance purposes for a very long time, these unmanned arial vehicles (UAV) are traditionally equipped with high tech cameras to capture images from high altitudes. Light weight drones are not capable of carrying such high-tech equipment which results in sample images of low resolution. Mounted camera and drone itself are not safe from weather either. Strong turbulence and lack of light can result in loss in data quality. While image processing using deep learning algorithms have only improved over last few years, all these factors mentioned above are responsible for such slow pace of development in this field.

This proposed project emphasized on obtaining more refined output from drone images using existing deep learning algorithms and techniques. YOLO algorithm serves well to this project as it's famous for its accurate and fast image processing. YOLOR have proved to be significantly better regarding object detection as it puts both implicit and explicit knowledge at use, hence obtaining a much-refined representation of information in the images **[3]**. This project will push these algorithms to their limits as the sample data would contain high resolution images with target areas potentially be smaller in size which would make it challenging to detect.

Aims and Objectives:

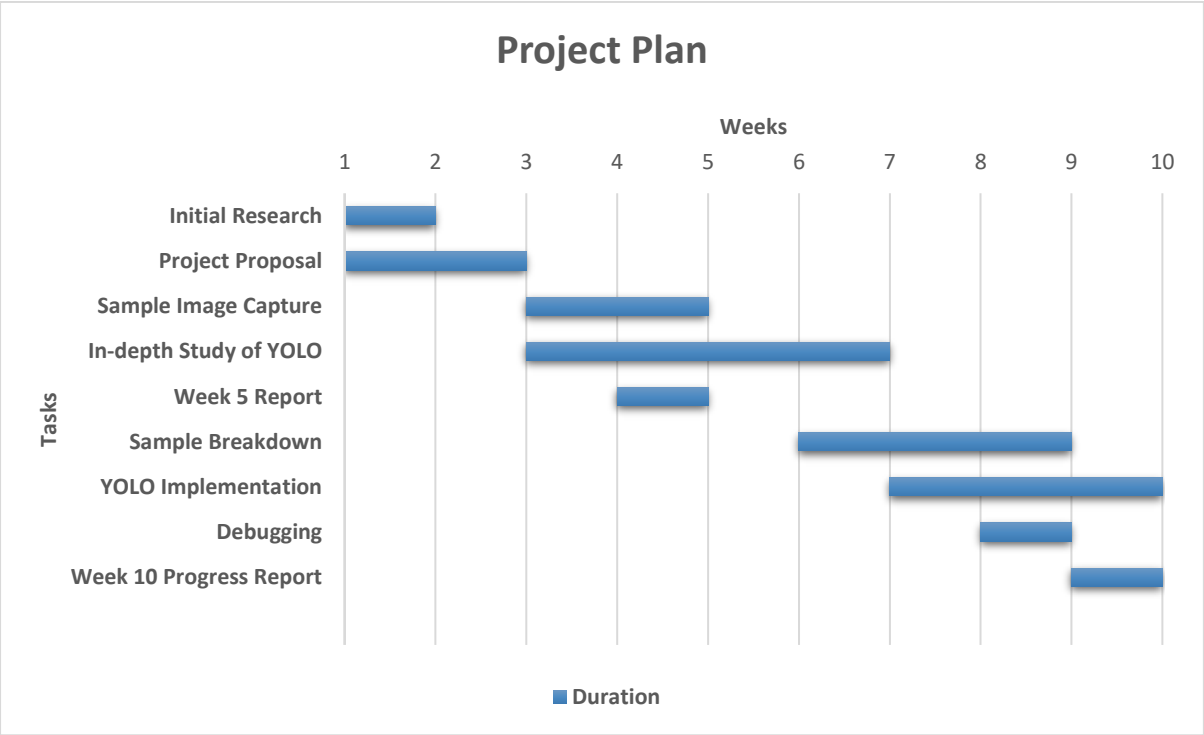
The aim of this project is to contribute toward tackling down waste crime using modern hardware and software. This refers to use of light weight compact drones with high quality camera and latest machine learning models and techniques. This project will be completing following objectives throughout development.

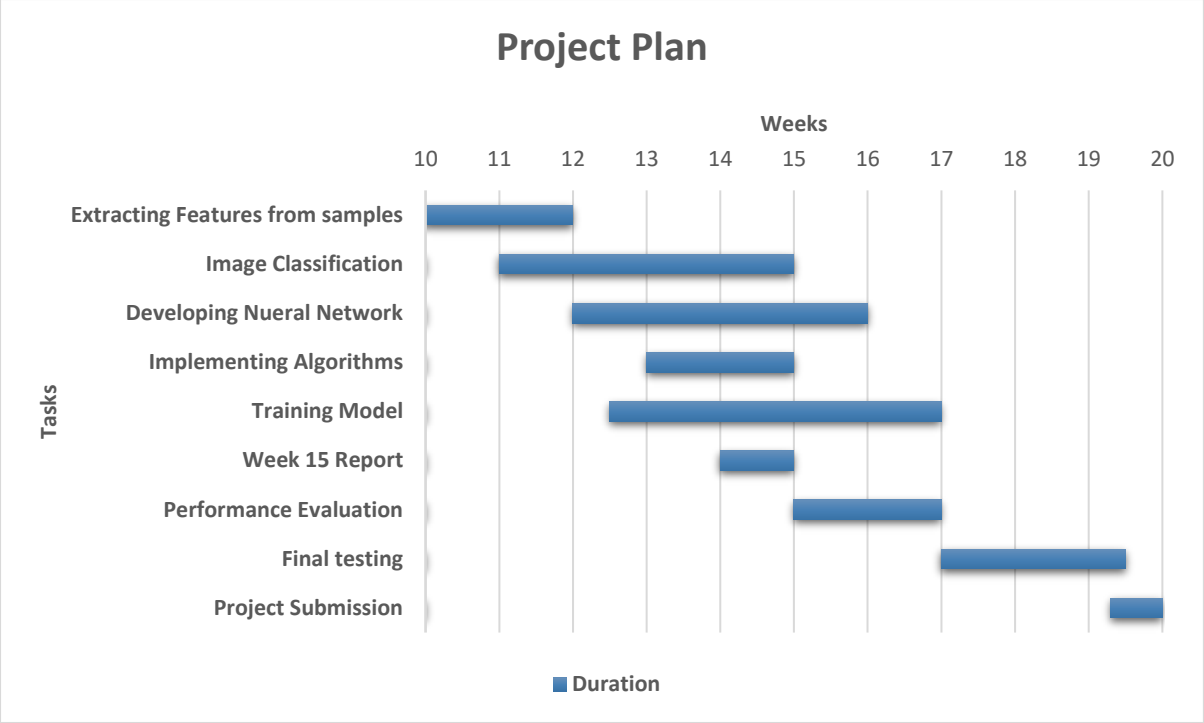
- Capturing sample data for research and development using drones
- Feature extraction and filtering sample data to train model with low costs
- Implementing YOLO algorithm in deep learning model
- Training model to detect waste found on residential or commercial areas
- Testing model with different techniques to find most effective way of tracing waste from drone images
- Analysis of the project to highlight findings that could contribute toward further development and improvements in this field.

Work Plan:

This proposal is the very first stage of this project. This proposal outlines all the goals and potential challenges it might face. After capturing satellite and drone images, next stage will be to understand fundamentals of algorithms used in these models. Raw input is processed into more meaningful datasets which also reduces the amount of data. Afterwards, development of the deep learning model will be initialised using YOLO algorithm. A progress report on project work will be submitted during week 5 of first term. Next three weeks will be spent working on algorithm implementation. Week 10 will mark end of first term. Hence, another progress report will be submitted that will outline all work done by that time.

Actual development of the model will begin by start of second term. At this stage, extracting features from sample data will be focused onto so that image samples with higher information can be used to train the model. Image classification and object localisation are the stages where the refined input will be further processed so that model can understand what objects to look for in the boundary boxes. By the end of week 15, another brief progress report will be submitted explaining all the objectives and milestones achieved. Next few weeks will be spent testing the model with sample data containing poor quality image samples to understand flaws and limitations of the model. Further work will be done in order to improve performance and output quality.




Project Plan
Weeks
10
11
12
13
14
15
16
17
18
19
20
Tasks
Extracting Features from samples
Image Classification
Developing Nueral Network
Implementing Algorithms
Training Model
Week 15 Report
Performance Evaluation
Final testing
Project Submission
Duration

References:

[1] Waste crime fask force's crackdown

https://www.gov.uk/government/news/successful-first-year-for-waste-crime-taskforce-despite-pandemic

[2] You Only Look Once: Unified, Real-Time Object Detection

https://paperswithcode.com/paper/you-only-look-once-unified-real-time-object

[3] You Only Learn One Representation: Unified Network for Multiple Tasks

https://paperswithcode.com/paper/you-only-learn-one-representation-unified